Boost System Performance with Linux Kernel Recompilation/Optimisation

Are you facing performance issues with your old system, but don't want to spend money buying a new one? The techniques mentioned in this article could boost your system's performance by just making a few minor changes in the Linux operating system.

Linux is growing by the day, through the development of yet more distributions and new features. It is being used by almost all companies, and these firms need good system performance to efficiently achieve their goals. Currently, Linux distributors provide generic operating systems for different types of computer hardware specifications. We get good performance when there is a combination of good hardware and software. In case hardware is costly for the system, we need to focus on software. The aim of this article is to increase the performance and booting time of a specific system, as well as to ensure kernel optimisation and proper utilisation of system resources.

To boost the performance of a system, we just need its hardware and software components to operate efficiently. Linux is an open source operating system and anyone can modify its core components. Linux distributors provide a generic operating system for all the hardware. In that OS, some packages are not useful for some users and unnecessarily take up system resources, reducing its performance. We can improve a system's performance by focusing on both the hardware and software components.

If we consider hardware resources, we may need to buy new, powerful hardware, which can be expensive and will make the old hardware useless. Not only will we incur additional costs, but also unnecessarily generate e-waste.

On the other hand, software is an important parameter for performance enhancement. Software coding improves hardware performance and removes unnecessary stuff from the system. Hardware can be optimised only up to a certain level to improve a system's performance. But with the use of the appropriate software, the performance levels of hardware can be increased and much greater performance can be extracted from it.

A word of caution here, though. Once the hardware is optimised (by using the right software) to operate at a higher level, the performance of the system will be enhanced but can affect the hardware. The hardware may heat up and, in the worst cases, this could damage the whole system.

Good performance = Good software + Good hardware

The following are the parameters on which the performance of the system depends:

1. CPU frequency and architecture, along with the number of cores
2. Cache memory
3. File system
   · EXT2
   · EXT3
   · ReiserFS
   · EXT4
   · BTRFS
4. Data structure
5. Kernel recompilation
6. Network parameters